Name:

Roll no:

AM2540
BUCKLING OF STRUTS

Objective: To study the buckling of a column under axially applied load with various boundary conditions and compare the experimental buckling loads with the Euler buckling formulae.

Apparatus:

Figure 1 shows the details of the Buckling of Struts apparatus. It consists of a back plate with a load cell at one end and a device to load the struts at the top. The bottom chuck fixes to an articulated parallelogram mechanism, which prevents rotation but allows movement in the vertical direction against the ring load cell. The mechanism reacts to the considerable side thrust produced by the strut under buckling conditions, with little friction in the vertical direction. It has a digital force display also.

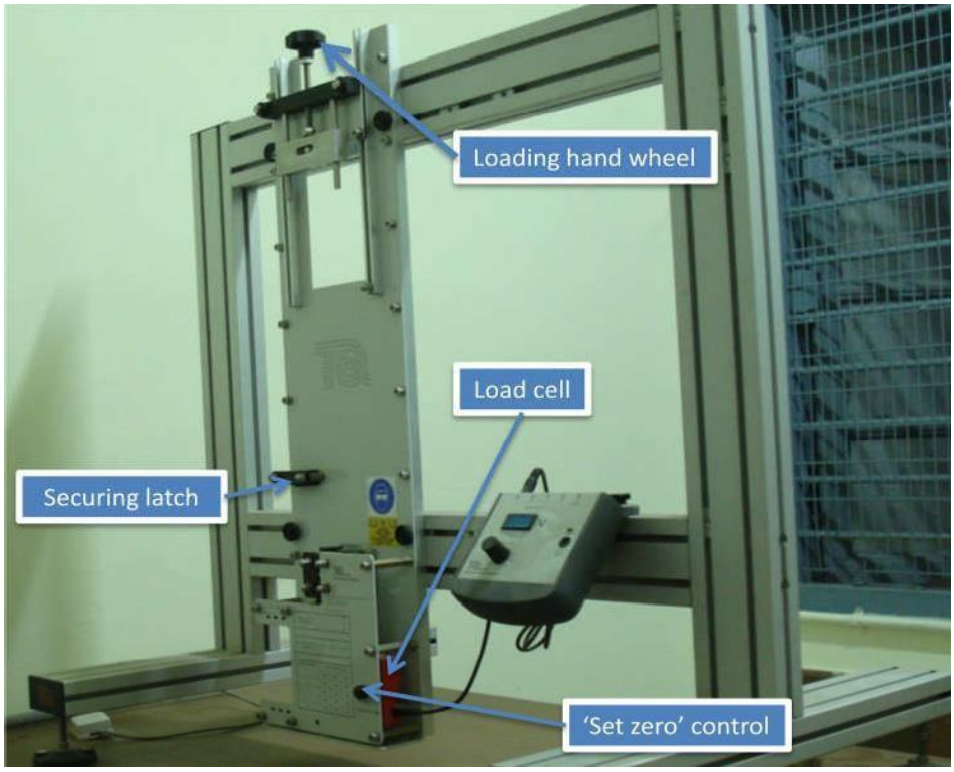


Fig. 1 Buckling of struts experimental setup

Theory:

The general Euler buckling formula for struts is

$$P_{cr} = \frac{\pi^2 EI}{L_e^2}$$

Where, Pcr is the Euler buckling load (N), E is Young's modulus (GPa), I is the second moment of area (mm^4). Choose appropriately I_{xx} or I_{yy} for given cross section) L_e = Effective length of strut (mm);

(For Hinged- Hinged L_e = L, Hinged-Fixed L_e=L /√2 and Fixed-Fixed L_e = L/2)

Young's modulus Steel= 210 GPa

Young's modulus Aluminium= 70 GPa

Procedure:

Make sure the digital force display is 'on'. Check that the mini DIN lead from 'Force Input 1' on the Digital Force Display to socket marked 'Force Output' on the right – hand side of the unit. Carefully Zero the force meter using the dial on the front panel of the instrument. Gently apply a small load with a finger to the top of the load cell mechanism and release. Zero the meter again if necessary. Repeat to ensure that the meter returns to zero.

Note: If the meter is only +/- 1N, lightly tap the frame (there may be a little 'stiction' and this should overcome it)

1. Buckling Load of a Pinned-Pinned End Strut:

i. Referring to Figure 3, fit the bottom chuck to the machine and remove the top chuck.
ii. Use the strut given for your experiments. Measure the cross section using the vernier.
iii. Adjust the position of the sliding cross crosshead to accept the strut using the thumbnuts to lock off the slider. Ensure that there is maximum amount of travel available on the hand wheel thread to compress the strut. Finally tighten the locking screws.
iv. Carefully back off the hand wheel so that the strut is resting in the notch but not transmitting any load; rezero the forcemeter using the front panel control.
v. Carefully start the loading of the strut. If the strut begins to buckle to the left, "flick" the strut to the right and vice versa (this reduces any errors associated with the straightness of the strut)

vi. Turn the hand wheel until there is no further increase in load (the load may peak and then drop as it settles into the notches). *Do not load the struts after the buckling load has been reached otherwise the strut will become permanently deformed.* Record the final load as shown in sample table under 'buckling load'. Try loading the strut three times and take the average of these as the experimental buckling load.



Figure 2 Experimental layout (Pinned-Pinned ends)

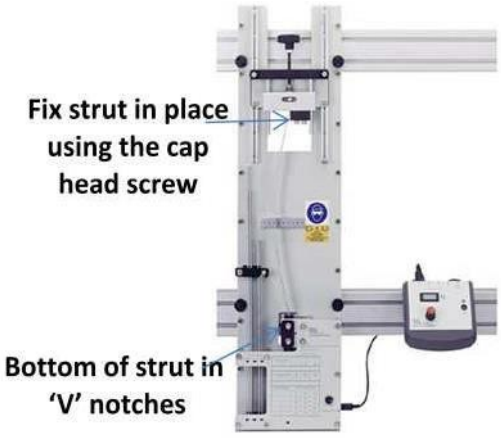


Figure 3 Experimental layout (Fixed-Pinned ends)

2. Buckling load of a Pinned – Fixed End Strut

i. Follow the same procedure as in experiment 1, but this time remove the bottom chuck and clamp the specimen using the cap head screw and plate to make a pinned –fixed end condition (Fig. 4). Do not load the struts after the buckling load has been reached otherwise the strut will become permanently deformed.

ii. Record your results as per the sample table. Note that the test length of the struts is shorter than in Experiment 1 due to allowance made for clamping the specimen. Use this changed length in calculations.

Observations:

1. Buckling Load of a Pinned-Pinned End strut

Strut No.	Length (L) mm	Thickness (d) mm	Breadth (b) mm	I_{xx} (mm^4)	I_{yy} (mm^4)	Buckling load (P_{cr}) (Experimental) N		Buckling Load theory N	% Error
						Trial	Average		
1 .a 1.b 1.c 1.d									

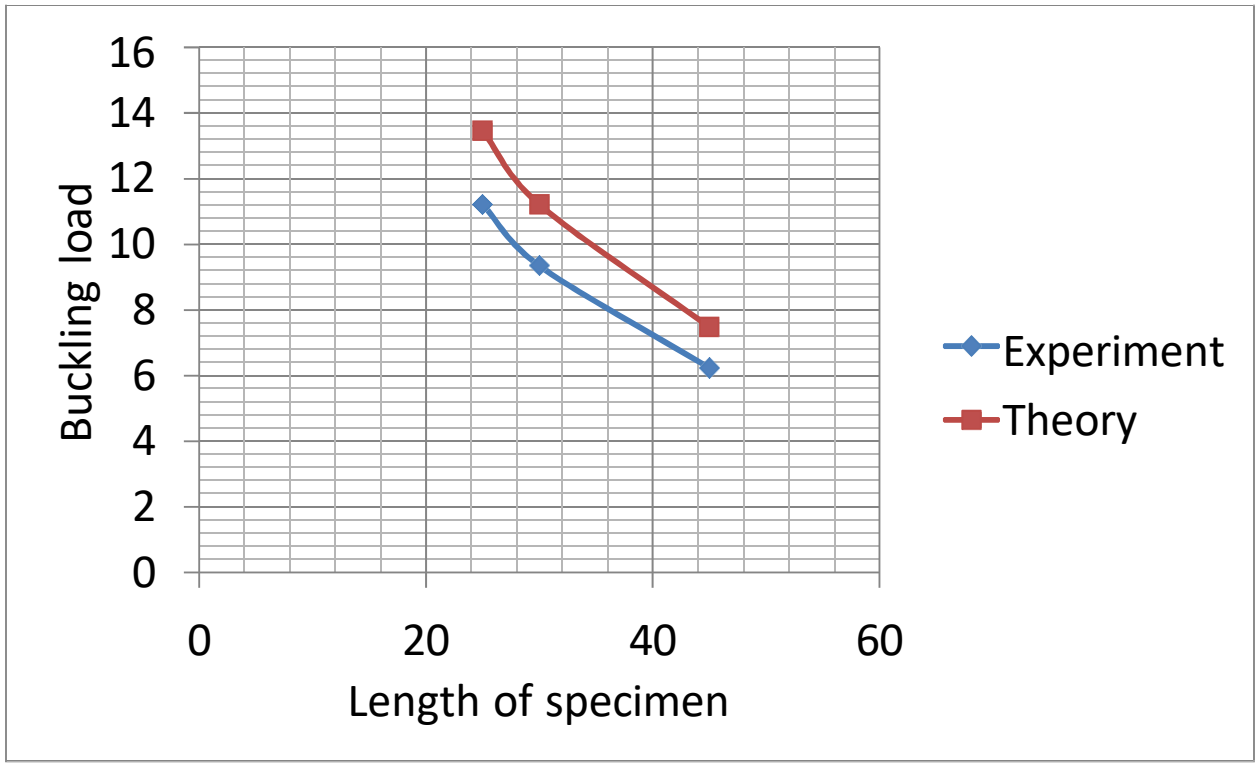
2. Buckling Load of a Pinned - Fixed End Strut

Strut No.	Length (L) mm	Thickness (d) mm	Breadth (b) mm	I_{xx} (mm^4)	I_{yy} (mm^4)	Buckling load (P_{cr}) (Experimental) N		Buckling Load theory N	% Error
						Trial	Average		
2 .a 2.b 2.c 2.d									

Graph:

Separate print out of the graph (sample as shown below) is to be attached along with the report.

Buckling load Vs Length



Results:

Conclusions:

Questions

1. How do imperfections in a column (such as slight bends, initial crookedness, or variations in material properties) affect the actual buckling load compared to the theoretical value predicted by Euler's formula?

2. Consider the design of railway tracks exposed to extreme temperature variations. Explain why tracks may buckle under certain conditions.

3. Why does slenderness ratio play such a significant role in the susceptibility of columns to buckling, and how would you optimize column geometry and material selection in a high-rise building to avoid both elastic and inelastic (yielding) buckling?

4. Suppose your experimental results for the critical buckling load of a pinned-pinned strut are consistently lower than the theoretical values given by Euler's formula. What factors or sources of error could explain this discrepancy? How would you experimentally validate your hypothesis about these sources

5. Imagine you conducted the experiment with struts made from different materials (e.g., steel, aluminum, brass) but with identical dimensions and end conditions. Based on the theory, how should their buckling loads compare? Discuss why the experimental results might deviate from this expectation and propose material selection strategies for structural columns in practice

6. During the experiment, if you kept the length of the strut constant but altered the end conditions (pinned-pinned, fixed-fixed, fixed-free, etc.), how would the critical buckling load change? Explain why, using theory and your own experimental data. How can this inform design choices in structures exposed to varying boundary conditions?

7. Propose a modification to the standard buckling experiment that would allow you to systematically study the effect of initial imperfections or misalignments in the strut. What measurements would you take, and how might the results influence safety factors in structural design